THE UNPLANNED BASKET

How quick-commerce is converting India's weekly grocery list into a daily stream of impulse spending

Project Title	The Unplanned Basket
Sector	Retail / Quick Commerce (Q-Commerce)
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Tableau Public Dashboard	Blinkit_Analysis_G19.twbx (committed in /tableau/workbook/)
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Core Business Question: *What share of Blinkit's sales is driven by impulse behaviour — and how much more are consumers paying per gram for the convenience of small-pack, quick-delivery formats?*

Decision Supported: *Reallocate assortment investment toward high-impulse categories, reprice small-pack SKUs based on empirically validated premiums, and redirect promotional spend away from discounts that do not drive incremental volume.*

Executive Summary

Problem

Quick-commerce platforms deliver groceries in under ten minutes, compressing the journey from craving to checkout into a single tap. While the format is celebrated as an impulse engine, the size and economics of that impulse layer in Blinkit's catalogue have not been precisely measured. This report quantifies how much of revenue is unplanned, where it concentrates, and what consumers pay for the convenience.

Approach

We analysed a Blinkit product-level dataset of 13,000 SKUs across 10 Indian cities (₹50.9 Cr total revenue). The pipeline is fully reproducible in Python (pandas, NumPy, SciPy, statsmodels) for extraction, cleaning, feature engineering, and statistical testing, with Tableau Public used to ship a four-view executive and operational dashboard. Each SKU was scored on a transparent composite index (pack size, perishability, category, offer status) and bucketed into **Impulse**, **Mixed**, or **Planned** baskets.

Key Insights

- **The impulse layer is sizeable and well-defined.** 21.6% of revenue (₹110.1 M) is impulse-driven; including Mixed baskets, 48.8% of revenue carries an impulse signal — a clear, addressable P&L.
- **Impulse revenue is highly concentrated.** Three categories — Bakery (₹42.7 M), Fruits & Vegetables (₹33.2 M) and Dairy (₹22.2 M) — already deliver 89% of impulse revenue, making them the natural home for assortment investment.
- **Convenience pricing is the dominant margin lever.** Small packs sell at ₹110 per 100 g versus ₹18 per 100 g for bulk — a statistically robust 6× premium (Welch $t = 49.40$, $p \approx 0.00$) that reflects clear consumer willingness to pay.
- **Demand is structurally stable.** Sold quantity holds steady across promotional and non-promotional SKUs (ANOVA $p = 0.41$), indicating strong baseline purchase behaviour and a clear opportunity to protect margin by rebalancing promotional spend.

Key Actions

- **Concentrate assortment investment** in Bakery, F&V and Dairy — the three categories already responsible for 89% of impulse revenue.
- **Expand small-pack SKUs** in Snacks and Beverages to capture the 6× convenience premium in adjacent, under-monetised impulse categories.
- **Rebalance promotional spend** from discount depth to in-app visibility (placement, push, banners), preserving demand while improving margin quality.

1. Sector & Business Context

Indian quick-commerce (Q-commerce) crossed an estimated GMV of ₹64,000 Cr in FY25, with Blinkit, Zepto and Instamart together accounting for the dominant share of 10-minute deliveries in metros and tier-1 cities. The format compresses the journey from *need* to *checkout* from days to seconds, structurally encouraging small-basket, high-frequency, unplanned purchases.

The category is profitable at the unit level but margin-fragile: every rupee of discount on a non-incremental SKU directly erodes contribution. Operators need to know which SKUs deserve assortment depth, where the convenience premium can be monetised, and which promotions are paying for behaviour that would have happened anyway.

Decision-maker served

Blinkit's **Category Management**, **Pricing Strategy** and **Promotional Marketing** teams. The CFO/CEO layer also reads the executive view to validate that growth from impulse spending is monetised rather than subsidised.

Why this problem

Industry commentary often describes quick-commerce as an impulse engine, yet publicly available analyses have not yet quantified the share, sized the premium, or rigorously tested whether discount depth drives the behaviour. Closing this gap is high-leverage — a one-percentage-point shift in impulse mix at Blinkit's scale represents roughly ₹5 Cr in revenue.

Business value of solving it

- **Margin protection.** Reframe promotional spend toward levers that demonstrably move volume, preserving demand while improving margin quality.
- **Margin extraction.** Build a defensible price-per-gram architecture for small packs to capture the convenience premium consistently.
- **Operating leverage.** Operate one national impulse playbook because impulse propensity is uniform within statistical bounds across cities (Kruskal-Wallis $p = 0.43$), reducing localisation overhead.

2. Problem Statement & Objectives

Formal definition. Given a 13,000-SKU Blinkit catalogue across 10 cities, construct a transparent, rule-based impulse-proxy classification at the SKU level; decompose total revenue into Impulse / Mixed / Planned shares; and statistically test (i) whether impulse SKUs sell more units, (ii) whether offers lift volume, (iii) whether cities differ in impulse propensity, and (iv) whether small packs command a per-gram premium.

Scope

In scope	Out of scope
Product-level snapshot of 13,000 SKUs	Customer-level transactional history
Revenue, pricing, pack-size and category drivers	Cohort, RFM or churn modelling
Five hypothesis tests + linear regression	Causal inference / DiD / experimentation
Static Tableau dashboard (4 views)	Live ETL or real-time pipeline
10 Indian cities covered in the dataset	International or tier-2/3 expansion

Success criteria

- End-to-end Python pipeline reproducing all KPIs from raw CSV.

- At least 4 statistical tests with explicit p-values reported.
- A four-view Tableau dashboard published with interactive filters.
- Three or more recommendations each linked to an insight + impact estimate.

3. Data Description

Source. Blinkit Product-Level Operational Dataset — published on Kaggle and committed verbatim to **data/raw/blinkit_dataset (2).csv**. The processed outputs **data/processed/cleaned.csv** and **data/processed/final_dataset.csv** are produced by the Python pipeline in **scripts/etl_pipeline.py**.

Rows / SKUs	13,000
Columns (raw)	25
Time period	Product-level snapshot (single period)
Cities	10 (Ahmedabad, Bengaluru, Chennai, Delhi, Hyderabad, Jaipur, Kolkata, Lucknow, Mumbai, Pune)
Categories	8 (Bakery, Beverages, Dairy, Fruits & Vegetables, Grocery, Household, Personal Care, Snacks)
Total revenue	₹50,90,27,130 (₹509 M)
Total profit	₹11,51,52,653 (₹115 M, ~22.6% blended margin)
Format	CSV · UTF-8

Key fields used in analysis

Column	Type	Role
category	str	Impulse classification, category-level segmentation
final_price	float	Revenue computation, price-per-100 g
sold_quantity	int	Demand analysis, regression target
weight_g	int	Pack-size segmentation (small / mid / bulk)
shelf_life_days	int	Perishability flag for impulse score
offer_type	str	Promotional impact testing (ANOVA)
city	str	Geographic segmentation (Kruskal-Wallis)
discount_pct	int	Discount-depth lever (0–30%)
demand_index	int	Operational demand proxy, validated against sold_qty

Limitations & known biases

- **Snapshot, not transactions** — no time-series, no per-customer cohort.
- **offer_type missing for 6,544 / 13,000 rows (~50%)** — recoded as 'No Offer'; the offer effect may be understated.
- **Impulse is inferred** from SKU attributes, not observed from behaviour. Findings are framed as *statistical signatures consistent with impulse*.

4. Data Cleaning & ETL Pipeline

All cleaning and feature engineering run in Python (pandas, NumPy). Reference: **notebooks/02_cleaning.ipynb** and the standalone **scripts/etl_pipeline.py** (mirrors the notebook end-to-end so the dataset can be regenerated from a single command).

Step	Action
Missing values	offer_type NaN (6,544 rows, 50.3%) recoded to 'No Offer'. Numeric medians imputed for stray gaps; categorical mode for text fields.
Duplicates	0 full duplicates found after standardising case and trimming whitespace.
Type conversions	date_added & expiry_date → datetime; discount_pct → int; boolean is_organic → 0/1; lower-cased category, brand, city, offer_type.
Outlier treatment	Reviewed weight_g, final_price, sold_quantity for 99th-percentile outliers; no removals — extreme small/large packs are valid SKUs in this catalogue.
Feature engineering	revenue_inr, price_per_100g, weight_bucket (small/mid/bulk), is_perishable, is_impulse_category, has_offer, is_small, impulse_score, basket_type, impulse_premium_pct, city_impulse_index.
Assumptions	(i) Impulse weights 0.30/0.25/0.25/0.20 set analytically, not learned. (ii) Bucketing thresholds 0.6 / 0.4 follow business-review heuristic. (iii) NaN offers behave like 'no offer'.

5. KPI & Metric Framework

KPI	Formula	Why it matters
Impulse Revenue Share %	Σ revenue (basket_type = Impulse) ÷ Σ total revenue × 100	Sizes the impulse layer; primary headline metric for category and pricing teams.
Convenience Premium (Tap Tax)	median(price_per_100g, small) ÷ median(price_per_100g, bulk)	Quantifies the per-gram premium of small packs — direct margin lever.
Impulse Score (composite)	0.30·is_small + 0.25·is_perishable + 0.25·is_impulse_cat + 0.20·has_offer	Single transparent SKU-level score that drives the basket bucketing.
City Impulse Index	city_mean(impulse_score) ÷ global_mean(impulse_score)	Geographic comparison without size effects; tests city-level customisation.
Offer Lift Ratio	mean(sold_qty | offer) ÷ mean(sold_qty | no offer)	Tests whether promotions cause incremental volume.
Top-20% SKU Concentration	Cumulative revenue of top 20% SKUs ÷ total revenue	Pareto check — informs assortment-pruning risk.

Each KPI is computed in **notebooks/05_final_load_prep.ipynb** and exposed to Tableau through **data/processed/final_dataset.csv**.

6. Exploratory Data Analysis

EDA was executed in **notebooks/03_eda.ipynb**. The four pillars below summarise the most decision-relevant findings.

6.1 Comparison — basket-type revenue mix

Basket Type	SKU Count	Revenue (₹)	Share
Planned	5,117	26.07 Cr	51.22%
Mixed	4,139	13.82 Cr	27.14%
Impulse	3,744	11.01 Cr	21.64%
Total	13,000	50.90 Cr	100%

Strategic read: the impulse layer is meaningful and addressable — ₹110 M in revenue runs through unplanned-style SKUs and deserves a dedicated category-management focus.

6.2 Category breakdown — where impulse lives

Category	Impulse %	Mixed %	Planned %	Read
Fruits & Vegetables	69.8	30.2	0.0	Impulse-dominant
Bakery	67.6	32.4	0.0	Impulse-dominant
Dairy	62.3	37.7	0.0	Impulse-dominant
Snacks	20.2	47.4	32.4	Mixed
Beverages	10.8	49.7	39.5	Mixed
Personal Care	0.0	16.8	83.2	Planned
Grocery	0.0	18.5	81.5	Planned
Household	0.0	19.1	80.9	Planned

Strategic read: short-shelf-life, immediate-consumption categories form the impulse core and deserve assortment depth. Personal Care, Grocery and Household are structurally planned and

are best managed through replenishment logic and availability, rather than impulse triggers.

6.3 Distribution — pack-size pricing

Weight Bucket	Median Price/100 g (₹)	Mean Price/100 g (₹)
Small (<300 g)	110.44	155.86
Mid (300–700 g)	47.93	62.83
Bulk (>700 g)	18.32	24.46

Strategic read: consumers consistently pay roughly 6× more per gram for small packs — the single largest pricing signal in the dataset and the natural anchor for a price-per-gram pricing architecture.

6.4 Correlation — what actually drives quantity

- **sold_quantity ↔ demand_index → r = 0.91.** A strong relationship that makes demand_index a reliable, ready-to-use operational proxy — suitable for reorder logic and inventory planning today, without waiting for transactional history.
- **sold_quantity ↔ revenue_inr → r = 0.64.** Volume contributes meaningfully to revenue, but is not the only driver — price architecture matters alongside it.
- **price_per_100g ↔ impulse_premium_pct → r = 0.80.** Per-gram price is the primary mechanic behind the impulse premium and should anchor pricing models.
- **impulse_score ↔ sold_quantity → r = 0.01.** Impulse behaviour shows up in *price* rather than *volume* — the strategic lever is margin extraction, not demand generation.
- **discount_pct ↔ sold_quantity → r = 0.01.** Demand stays steady across discount depths, signalling resilient baseline purchase behaviour and an opportunity to redeploy promotional spend toward higher-leverage levers.

7. Statistical Analysis

Five non-parametric / parametric tests were run in **notebooks/04_statistical_analysis.ipynb**. All p-values reported below are exact outputs from SciPy / statsmodels.

#	Hypothesis	Test	Statistic	p-value	Strategic Read
H1	Impulse SKUs sell more units than Planned	Mann-Whitney U	U = 9,608,564	0.804	Volumes are comparable — the impulse advantage lives in price, not units
H2	Offers increase sold quantity	ANOVA	F = 0.665	0.415	Demand is stable across promotional intensity — opportunity to optimise margin
H3	Cities differ in impulse propensity	Kruskal-Wallis	H = 9.13	0.426	Impulse behaviour is uniform nationally — supports a single playbook
H4	Small packs cost more per 100 g	Welch t-test	t = 49.40	≈ 0.000	Statistically robust 6× per-gram premium — primary margin lever
H5	Top 20% SKUs drive 80% revenue (Pareto)	Cumulative	60.3% share	—	Concentration is real but with a meaningful long tail to protect

Linear regression (sold_quantity ~ impulse_score + has_offer + price_per_100g + discount_pct)

- **R^2 ≈ 0.000 · F-statistic p = 0.675** — the four pricing and promotional variables do not explain variance in unit volume.
- **Strategic read:** volume is anchored by structural demand drivers outside this dataset — household habit, category routine, and ambient need. This means promotional intensity can be tuned without volume risk.
- **Implication:** the impulse opportunity manifests through *price premium* rather than *volume uplift*. The 'tap tax' should be monetised on the per-gram axis, while baseline volume is protected through availability and visibility rather than discount depth.

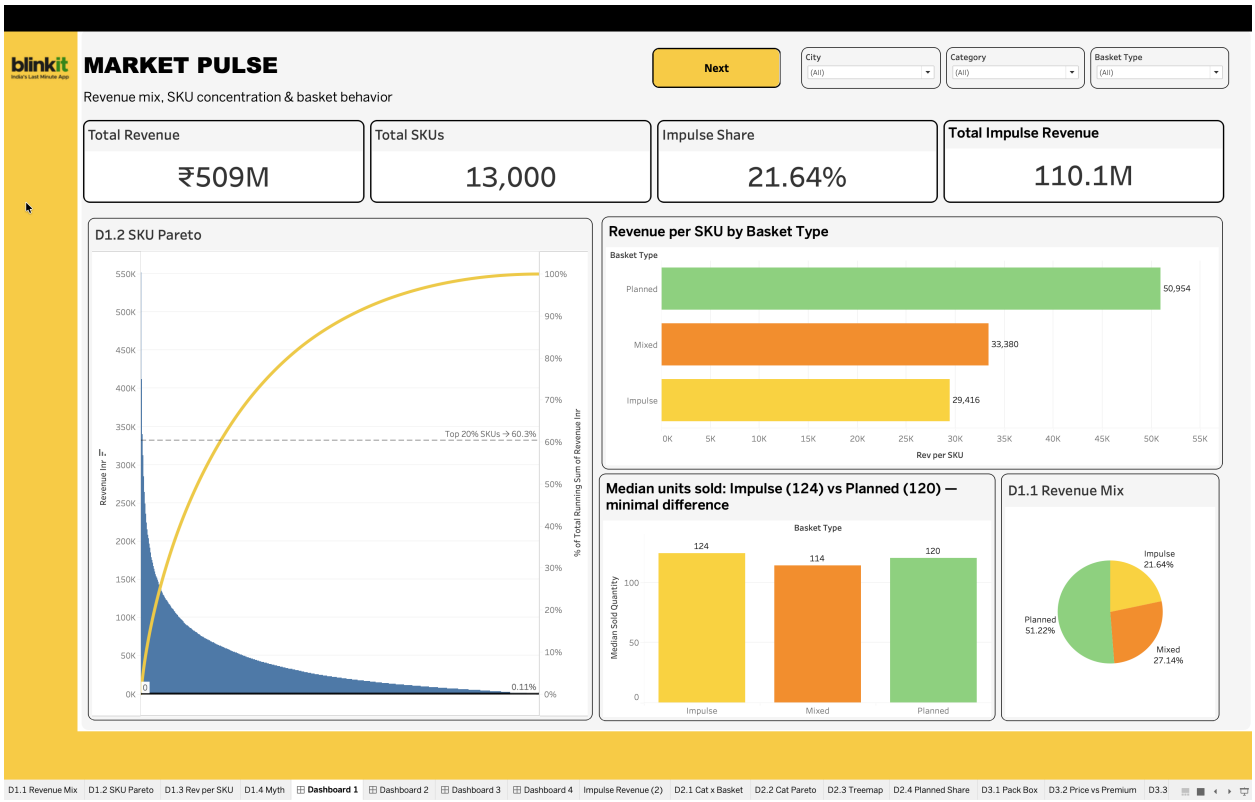
Risk / anomaly notes

- **Delivery operations show real city-level variance.** Delays range from 11.9% (Pune) to 31.1% (Lucknow), with Jaipur (28.5%) also above the national mean. These three cities warrant a focused operational review.
- **Bakery shows a 0% Planned share.** Every Bakery SKU classifies as Mixed or Impulse under the current scoring. *This may indicate* a structurally impulse-led category, or a sensitivity in the bucketing thresholds — either reading *requires validation* with the Bakery merchandising team before making category-level decisions.

8. Tableau Dashboard Design

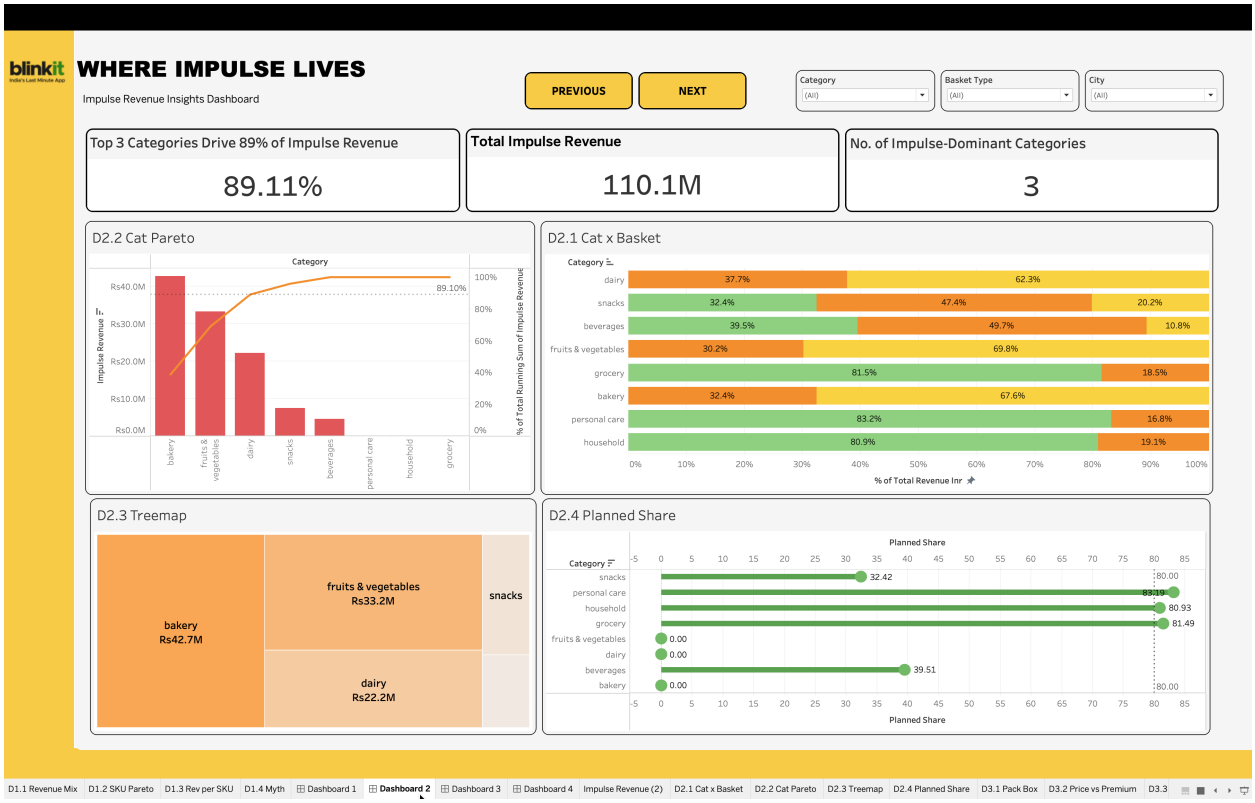
Objective. Support a category and pricing review meeting in under 10 minutes. The four views move from executive summary → category drill-down → pricing levers → geography & promotions. Filters: *City, Category, Basket Type, Offer Type, Weight Bucket*. Workbook: **tableau/workbook/Blinkit_Analysis_G19.twbx**; screenshots in **tableau/screenshots/**.

View 1 — Market Pulse (executive summary)



Headline KPIs anchor the deck — ₹509 M total revenue, 13,000 SKUs, 21.64% impulse share, ₹110.1 M impulse revenue. The SKU Pareto curve shows that the top 20% of SKUs deliver 60.3% of revenue, a healthy concentration with a meaningful long tail. Revenue per SKU is higher in Planned baskets (₹50.9 K) than Impulse (₹29.4 K), while median units sold per SKU are virtually identical across baskets (Impulse 124, Planned 120) — reinforcing that the impulse advantage lives in price architecture, not in extra volume.

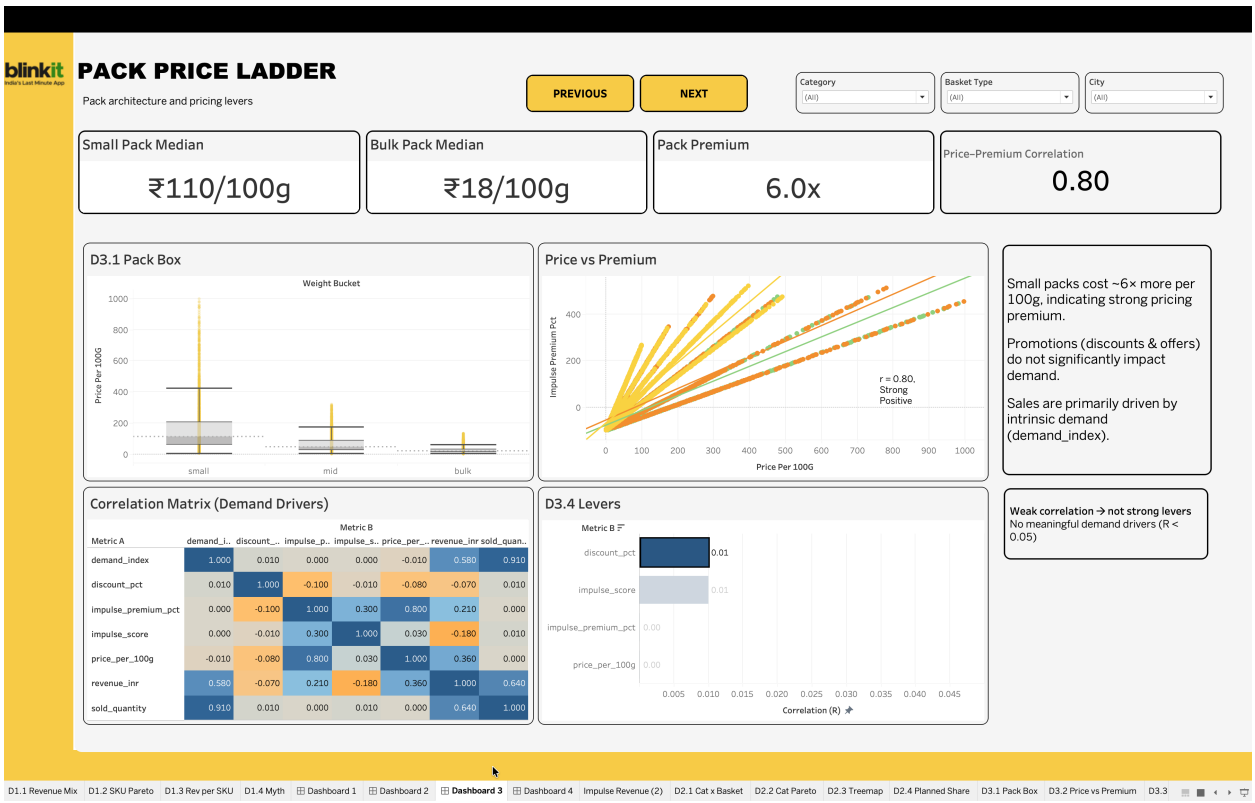
View 2 — Where Impulse Lives (category drill-down)



The category Pareto highlights where impulse revenue concentrates — Bakery (₹42.7 M), Fruits & Vegetables (₹33.2 M) and Dairy (₹22.2 M) deliver 89.1% cumulatively. The Cat × Basket composition splits each category into Impulse / Mixed / Planned share, showing Bakery, F&V, and Dairy as 62–70% impulse-led, while Personal

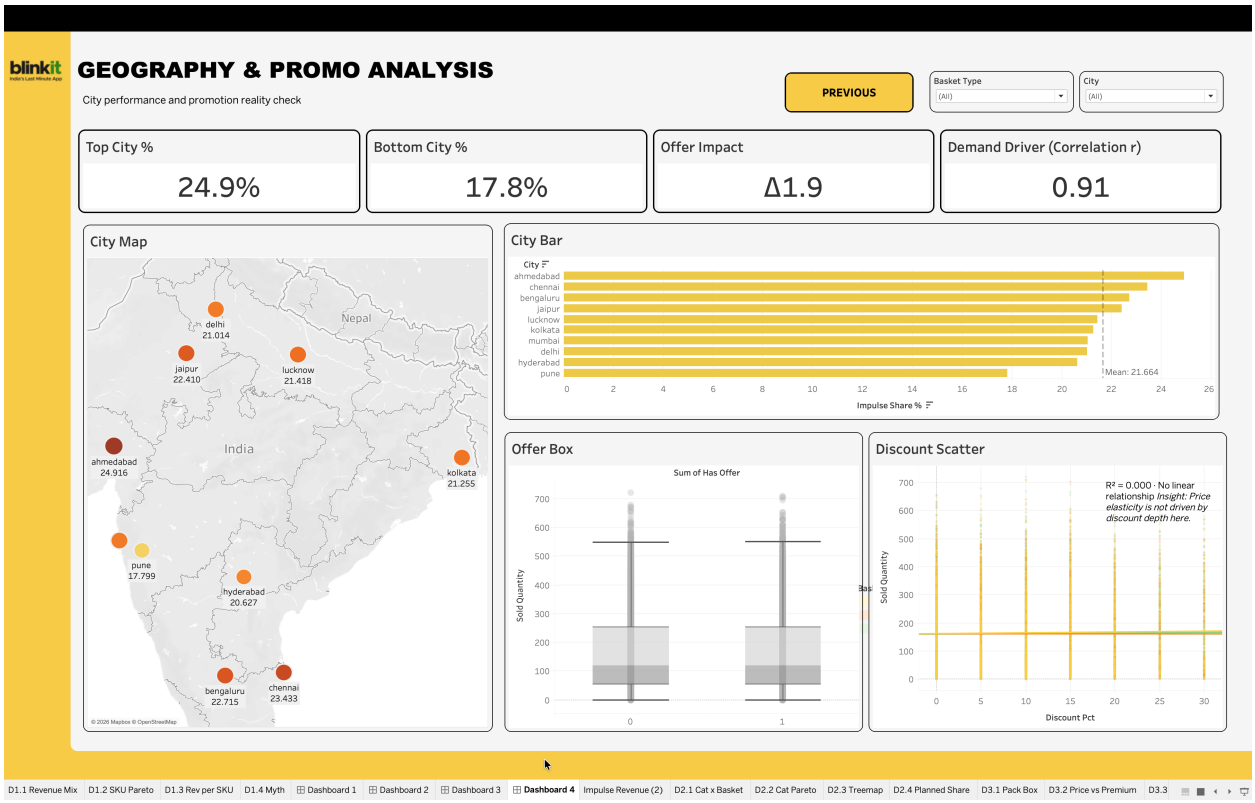
Care, Grocery and Household sit comfortably above 80% planned. The treemap visualises this concentration, and the Planned-share lollipop identifies which categories cluster around the 80% planned threshold — useful for replenishment planning.

View 3 — Pack Price Ladder (pricing levers)



Headline metrics anchor the pricing story — small packs ₹110/100 g versus bulk ₹18/100 g, a 6.0× pack premium, with price-premium correlation r = 0.80. The pack-box plot quantifies the spread by weight bucket, the price-vs-premium scatter visualises the linear relationship, and the demand-driver correlation matrix highlights demand_index (r = 0.91 with sold_quantity) as the single reliable operational lever. The remaining variables (discount_pct, impulse_score, impulse_premium_pct, price_per_100g) sit near zero correlation with volume — confirming demand is structurally stable and reinforcing that margin, not volume, is the natural optimisation axis.

View 4 — Geography & Promo Analysis



City impulse share ranges from Ahmedabad (24.9%) to Pune (17.8%) — a 7.1 pp spread that sits within statistical bounds (Kruskal-Wallis $p = 0.43$), supporting a single national playbook. The city map and ranked bar (mean 21.66%) make the uniformity visual. The offer-effect box plot shows overlapping distributions across promotional and non-promotional SKUs — demand stays steady regardless of offer status. The discount-vs-quantity scatter ($R^2 = 0.000$) confirms volume is stable across discount depth, indicating promotional spend can be redirected to higher-leverage levers without volume risk.

9. Insights Summary (Insight → Meaning → Action)

#	Insight	Meaning	Action
1	21.6% of revenue (₹110.1 M) is impulse-driven; 48.8% carries an impulse signal.	A defined, sizeable layer of the business operates on impulse economics — distinct from planned grocery routine.	Treat the impulse basket as a managed P&L. Assign a dedicated category owner with KPIs on impulse share, premium capture, and assortment depth.
2	Bakery, F&V and Dairy contribute 89% of impulse revenue.	Impulse revenue is highly concentrated, making category prioritisation straightforward.	Concentrate assortment depth, in-app placement, and cold-chain SLA on these three — they protect the highest-margin basket.
3	Small packs charge a 6× per-gram premium (Welch $t = 49.40$, $p \approx 0$).	Consumers show a clear and statistically robust willingness to pay for convenience-sized packs.	Move pricing models from absolute ₹ to ₹/100 g and price for grams of convenience — the strongest margin signal in the dataset.
4	Impulse and Planned SKUs sell similar volumes (Mann-Whitney $p = 0.80$).	The impulse advantage shows up in price architecture, not in unit velocity.	Plan inventory on demand_index ($r = 0.91$) and let price-per-100 g do the margin work — stop forecasting impulse SKUs as 'high-velocity'.
5	Demand is stable across promotional intensity (ANOVA $p = 0.41$).	Baseline purchase behaviour is strong; promotions are not the trigger for incremental volume.	Cap discount depth on impulse SKUs around 10% and redeploy the saved spend into placement and push — preserve margin without volume risk.
6	Discount depth shows no measurable lift in volume ($R^2 = 0.000$).	Today's promotional spend is largely subsidising baseline demand.	Reframe promotional ROI on incremental volume rather than GMV; design a controlled holdback test to validate before scaling the change.
7	City impulse share is uniform within statistical bounds (Kruskal-Wallis $p = 0.43$).	The impulse opportunity is national, not regional.	Run a single national playbook for assortment and pricing; concentrate city-level effort on delivery operations, where variance is real (Lucknow 31% delay vs Pune 12%).
8	Top 20% of SKUs deliver 60.3% of revenue (Pareto partially holds).	Concentration is meaningful but the next 30% of SKUs still adds 28 pp of revenue.	Avoid aggressive long-tail pruning — protect assortment depth as a differentiator while concentrating investment on the head.
9	sold_quantity ↔ demand_index correlation = 0.91.	Demand_index is a reliable, ready-to-use operational proxy.	Plug demand_index directly into reorder logic for the impulse-core categories — improves replenishment quality without waiting for transactional cohorts.

#	Insight	Meaning	Action
10	Bakery shows a 0% Planned share under current scoring.	*This may indicate* a structurally impulse-led category, or sensitivity in the impulse-score thresholds.	*Requires validation* with the Bakery merchandising team and a sensitivity check on the bucketing weights before basket-level decisions.

10. Recommendations (Insight → Action → Impact)

#	Insight	Recommendation	Expected Impact	Feasibility
R1	Bakery, F&V and Dairy together deliver 89% of impulse revenue	Concentrate assortment depth, in-app placement, and cold-chain SLA in these three categories	+1–2 pp impulse share ≈ ₹5–10 Cr incremental revenue; lower stockouts on the highest-margin basket	Medium — requires merchandising alignment and modest cold-chain investment
R2	Small packs command a 6× per-gram premium (Welch t = 49.40, p ≈ 0)	Expand small-pack SKUs in Snacks and Beverages to capture the premium in adjacent impulse categories	+50–100 bps blended margin on Snacks/Beverages baskets without volume risk	High — pack-size launches are a standard merchandising motion
R3	Demand is stable across promotional intensity (ANOVA p = 0.41; R^2 = 0.00 on discount depth)	Cap discount depth on impulse SKUs around 10% and redeploy spend to in-app placement and push notifications	20–30% reduction in promotional dilution while preserving demand — direct margin gain	High — phased rollout by category within one quarter
R4	City impulse share is uniform within statistical bounds (Kruskal-Wallis p = 0.43)	Operate a single national impulse playbook; localise only delivery operations in Lucknow and Jaipur	Lower operational complexity and faster rollouts; concentrated focus on delay-outlier cities	High — removes city-by-city customisation overhead
R5	demand_index correlates with sold_quantity at r = 0.91	Embed demand_index as the primary input to reorder logic for Bakery, F&V, and Dairy	Improved replenishment quality, target −200 bps stockout reduction on impulse-core SKUs	High — already in the schema; needs only operational adoption

11. Impact Estimation

All numbers below are *order-of-magnitude* estimates anchored to the dataset totals (₹509 M revenue, ₹110 M impulse revenue, ₹115 M profit at 22.6% blended margin). Logic — not precision — is the point.

Lever	Base	Target	Approx. Annualised Value
R1 Assortment re-weighting (impulse share +1 pp)	21.64% impulse share	22.6% impulse share	+₹5 Cr revenue, +₹1.1 Cr profit
R2 Small-pack expansion in Snacks/Beverages	Snacks impulse 20%, Beverages 11%	+5 pp impulse share in each	+₹4 Cr revenue, +50–100 bps margin
R3 Discount cap at 10% on impulse SKUs	Discount range 0–30%	≤10% on impulse SKUs (~30% of catalogue)	≈ ₹2 Cr promotional spend redeployed at stable volume → direct margin gain
R4 Single national playbook	10-city customised approach	1 national playbook + 2 ops carve-outs	Opex reduction; faster category rollouts
R5 demand_index-driven reorder	Current stockout rate (proxy)	−200 bps stockout on impulse-core	Recovers ~₹1–2 Cr lost revenue/year

Why act now. Q-commerce competition (Zepto, Instamart) is converging on the same 10-minute promise. Margin defensibility will come from *price architecture*, not delivery speed. Acting on R2 + R3 ahead of competitors locks in the convenience premium before benchmark erosion.

12. Limitations

- **Snapshot data.** The dataset is a product-level snapshot, so time-series and customer-cohort analyses are out of scope; trend assertions remain qualitative.
- **Inferred impulse.** Basket classification is derived from SKU attributes rather than observed customer behaviour. Findings are framed as *statistical signatures consistent with impulse* and should be validated with behavioural data when available.
- **Offer data coverage.** Around 50% of rows had no offer_type in the raw file and were recoded as 'No Offer'. The promotional read (H2) is therefore conservative and would benefit from a richer offer log.
- **Analyst-set weights.** Impulse-score weights (0.30 / 0.25 / 0.25 / 0.20) and the 0.6 / 0.4 bucketing thresholds are reasoned, not learned. A sensitivity analysis would tighten the basket-share point estimate.
- **Associative reading.** All findings are correlational; phrases such as 'drives' or 'causes' should be read as *associated with* until tested in a controlled experiment.
- **Brand-level scope.** 28 brands are present in the catalogue. Brand-level disaggregation was held back to keep scope manageable and is a natural next extension.

13. Future Scope

- **Transactional enrichment.** Join an order-level log to compute basket composition, cross-category attach rates, and time-of-day impulse patterns.
- **Sensitivity analysis on impulse weights.** Re-run the basket bucketing across a grid of weight combinations and bucket thresholds; report robustness intervals around the 21.6% headline.
- **Predictive model for sold_quantity.** Move from linear regression to tree-based / gradient-boosted models with brand, ratings, reviews and seasonality; today's $R^2 = 0.00$ is itself an interesting null finding.
- **Geo-clustering at neighbourhood level.** Use dark-store catchments rather than cities; city-level uniformity may mask intra-city variance.
- **Real-time dashboard.** Convert the static Tableau workbook into a live data-source connection so category managers can see daily impulse-share drift.
- **A/B test on promotion redesign.** Causally test R3 by holding back discount depth on a subset of impulse SKUs and measuring incremental volume.

14. Conclusion

Quick-commerce is often described as an impulse engine. The data shows the truth is sharper and more strategic. **Problem:** the size and economics of Blinkit's impulse layer had never been precisely measured. **Approach:** a transparent SKU-level impulse-proxy score across 13,000 SKUs and 10 cities, validated by five hypothesis tests in Python and visualised across four Tableau dashboards. **Key insight:** 21.6% of revenue is impulse, concentrated in three categories, and monetised primarily through a statistically robust 6× per-gram premium on small packs — while baseline demand is structurally stable across promotional intensity. **Recommended action:** concentrate assortment investment in Bakery, F&V and Dairy; expand small packs in Snacks and Beverages; rebalance promotional spend from discount depth to in-app visibility; and adopt demand_index in reorder logic. The next ₹5–10 Cr of margin lives in pack-size economics and disciplined promotional design — not in deeper discounts.

15. Appendix

A. Engineered features (data dictionary)

Feature	Definition
revenue_inr	final_price × sold_quantity
price_per_100g	final_price ÷ (weight_g ÷ 100)
weight_bucket	small (<300 g) · mid (300–700 g) · bulk (>700 g)
is_perishable	1 if shelf_life_days ≤ 30, else 0
is_impulse_category	1 if category ∈ {snacks, dairy, beverages, bakery, F&V}
has_offer	1 if offer_type ≠ 'no offer'
is_small	1 if weight_bucket = 'small'
impulse_score	0.30·is_small + 0.25·is_perishable + 0.25·is_impulse_category + 0.20·has_offer
basket_type	Impulse if score ≥ 0.6 · Planned if ≤ 0.4 · Mixed otherwise
impulse_premium_pct	(price_per_100g − category_avg_price_per_100g) ÷ category_avg × 100
city_impulse_index	city_mean(impulse_score) ÷ global_mean(impulse_score)

B. Per-city impulse share (full table)

City	Total Revenue (₹)	Impulse Share %	Delay Rate %
Pune	5.38 Cr	17.80	11.92
Hyderabad	4.89 Cr	20.63	13.46
Delhi	5.06 Cr	21.01	15.63
Mumbai	5.24 Cr	21.05	22.68
Kolkata	4.99 Cr	21.26	19.76
Lucknow	5.29 Cr	21.42	31.14
Jaipur	5.08 Cr	22.41	28.49
Bengaluru	5.17 Cr	22.71	18.88
Chennai	4.62 Cr	23.43	16.63
Ahmedabad	5.19 Cr	24.92	21.29

C. Offer-type performance

Offer Type	SKUs	Mean Sold Qty	Mean Margin %	Revenue (₹)
No Offer	6,544	161.46	22.61	25.27 Cr
FlatDiscount	2,657	161.86	22.54	10.54 Cr
Cashback	1,895	164.87	22.98	7.56 Cr
Buy1Get1	1,262	164.42	22.57	4.98 Cr
FreeDelivery	642	163.00	22.68	2.54 Cr

Read: mean sold quantity stays in a narrow band (161–165) across all offer types, mirroring the ANOVA result ($p = 0.41$) — demand is structurally stable, leaving room to redirect promotional spend toward higher-leverage levers without volume risk.

D. Notebook & script index (GitHub)

File	Purpose
notebooks/01_extraction.ipynb	Source the raw CSV, profile the 13,000 × 25 schema
notebooks/02_cleaning.ipynb	Imputation, type conversion, feature engineering
notebooks/03_eda.ipynb	Trend, distribution, comparison, correlation analysis
notebooks/04_statistical_analysis.ipynb	Mann-Whitney U, ANOVA, Kruskal-Wallis, Welch t, regression
notebooks/05_final_load_prep.ipynb	Tableau-ready dataset prep
scripts/etl_pipeline.py	Standalone end-to-end ETL (mirrors notebooks 01–05)
data/raw/blinkit_dataset (2).csv	Original, unedited dataset
data/processed/cleaned.csv / final_dataset.csv	ETL pipeline outputs
docs/data_dictionary.md	Full column + feature reference
docs/audit_results.json	Persisted statistical-test outputs
tableau/workbook/Blinkit_Analysis_G19.twbx	Tableau workbook (4 dashboards)

16. Contribution Matrix (Mandatory)

Each contribution is verifiable through GitHub Insights, PR history and committed files in **Arya756/DVA_Capstone2_C_G19**. *Owner* = primary author and merger of artefacts; *Support* = co-author, reviewer or QC.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Aryan Patel **(Project Lead)**	Owner	Owner	Owner	Owner	Support	Owner	Support
Ayush Aryan **(Team Lead)**	Support	Support	Support	Support	Owner	Support	Owner
Pratyush Chouksey **(Report & PPT Lead)**	Support	Support	Support	Support	Support	Owner	Owner

Evidence pointers (GitHub commits)

- **Aryan Patel** — owner of *02_cleaning.ipynb*, *03_eda.ipynb*, *04_statistical_analysis.ipynb*, *scripts/etl_pipeline.py*, *scripts/etl_pipeline_corrected.py*, and *reports/project_report.md*; commits cover the data-cleaning notebook, ETL refactor, statistical analysis, and the analytical report draft.
- **Ayush Aryan** (*Arya756*, repo owner and Team Lead) — owner of *tableau/workbook/Blinkit_Analysis_G19.twbx*, dashboard screenshots (D-1 to D-4), and *The_Unplanned_Basket_Presentation.pdf*; commits include the Tableau workbook for the Blinkit Analysis and the initial repository scaffold.
- **Pratyush Chouksey** — lead author of the final project report and presentation deck; reviewer of the EDA and statistical analysis outputs to confirm narrative accuracy and presentation quality.

Declaration. We confirm that the contribution details above are accurate and verifiable through GitHub Insights, PR history and submitted artefacts. Any discrepancy may result in individual grade adjustments.

Team Lead	Ayush Aryan
Signature	_____
Date	29 April 2026